

Capstone Proposal

Introduction

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Introduction

The eastern oyster (*Crassostrea virginica*) are viewed as a foundational species as they are directly responsible for creating and providing numerous ecosystem goods and services such as food and habitat production, shoreline stabilization and protection, water quality improvement, nutrient cycling, and carbon sequestration (Geselbracht et al. 2024; Grabowski et al. 2012). Additionally, they provide numerous economic benefits such as supporting oyster fisheries, recreational harvesting, tourism, along with the associated employment that goes along with those industries (Geselbracht et al. 2024; Peyre, Marshall, and Sable 2021). In one study, Grabowski et al. (2012) estimated the average annual value to range from \$10,325 to \$99,421 per hectare (2011 Dollars).

In recent decades, oyster reefs have experienced substantial declines, with only an estimated 15% remaining worldwide, and 50 to 80% being lost within the Gulf of (America) alone (Pollack et al. 2012; Peyre, Marshall, and Sable 2021). The decline is largely attributed to a combination of anthropogenic activities such as over harvesting, habitat alteration, sedimentation and water quality decline, disease and changing hydrology (Peyre, Marshall, and Sable 2021; Geselbracht et al. 2024). In response to these sharp declines, many restoration efforts have been undertaken with varying strategies and success rates; recently, planning documents released by the Deepwater Horizon Natural resource Damage Assessment Trustees called for largescale development of reef networks to increase resilience (Geselbracht et al. 2024).

2. Oyster decline and restoration within the Pensacola Bay System

1. Introduce the Pensacola Bay System (PBS) as the geographic focus of the study.
 - Describe PBS as a large estuarine system in northwest Florida.
 - Briefly describe its historical importance for Eastern oyster habitat.
 - Citation: (Geselbracht et al. 2024)
2. Describe the magnitude of oyster habitat loss within PBS.

- Contrast historical oyster reef extent with recent mapped oyster reef extent.
 - Establish the need for restoration and improved restoration-site selection.
 - Citation: (Geselbracht et al. 2024)
3. Explain why restoration-site selection is important.
- Restoration resources are finite.
 - Environmental conditions vary spatially throughout the estuary.
 - Poor site selection can reduce oyster survival, recruitment, and long-term reef persistence.
 - Citations: (Pollack et al. 2012; Barnes et al. 2007; Geselbracht et al. 2024)

3. Environmental controls on Eastern oyster habitat

1. Oyster occurrence and persistence are controlled by interacting physical, chemical, and biological conditions.
 - Establish habitat suitability as a multivariate problem.
 - Citations: (Peyre, Marshall, and Sable 2021; Pollack et al. 2012; Barnes et al. 2007)
2. Salinity is a major determinant of Eastern oyster distribution and habitat suitability.
 - Influences physiology, recruitment, disease, predation, and survival.
 - Represents an important environmental gradient within estuarine systems.
 - Citations: (Peyre, Marshall, and Sable 2021; Pollack et al. 2012; Barnes et al. 2007; Geselbracht et al. 2024)
3. Additional important habitat variables include dissolved oxygen, temperature, depth, turbidity, and substrate.
 - Dissolved oxygen influences physiological stress and mortality.
 - Temperature affects growth, reproduction, and physiological stress.
 - Depth influences physical habitat availability and environmental conditions.
 - Turbidity and sedimentation may influence feeding and reef persistence.
 - Suitable substrate is required for settlement and reef formation.
 - Citations: (Peyre, Marshall, and Sable 2021; Pollack et al. 2012; Barnes et al. 2007; Geselbracht et al. 2024)
4. Environmental predictors used in this study will therefore be selected based on both ecological relevance and available spatial data coverage.
 - Not every ecologically important parameter will necessarily provide adequate spatial data for predictive modeling.
 - Citations: (Peyre, Marshall, and Sable 2021; Geselbracht et al. 2024)

4. Habitat Suitability Indices as restoration decision-support tools

1. Introduce Habitat Suitability Indices (HSIs).
 - Describe HSIs as methods that transform environmental characteristics into standardized estimates of habitat suitability.

- Explain their use as spatial decision-support tools for restoration.
 - Citations: (Barnes et al. 2007; Pollack et al. 2012; Geselbracht et al. 2024)
2. HSIs have been applied specifically to Eastern oyster restoration.
 - Caloosahatchee Estuary, Florida.
 - Mission-Aransas Estuary, Texas.
 - Pensacola Bay System, Florida.
 - Citations: (Barnes et al. 2007; Pollack et al. 2012; Geselbracht et al. 2024)
 3. Describe the existing Pensacola Bay HSI.
 - GIS-based and estuary-specific.
 - Integrates biological, physical, chemical, and management-related factors.
 - Developed to identify candidate restoration areas within PBS.
 - Citation: (Geselbracht et al. 2024)
 4. Describe the strengths and limitations of HSI approaches.
 - Strengths include interpretability, adaptability, and integration of established ecological knowledge.
 - Results depend on decisions regarding variable selection, suitability functions, scoring, weighting, and aggregation.
 - Avoid characterizing HSI approaches as inherently inferior to statistical SDMs.
 - Citations: (Geselbracht et al. 2024; Johnson and Gillingham 2005)

5. Species distribution models as an empirical alternative or complement

1. Introduce species distribution models (SDMs).
 - SDMs quantitatively relate known species distributions to environmental conditions.
 - They can be used for ecological explanation and spatial prediction.
 - Citation: (Elith and Leathwick 2009)
2. Different habitat-modeling approaches can produce different spatial predictions.
 - Model selection and environmental-variable selection can influence areas identified as suitable.
 - Comparing modeling approaches can reveal areas of agreement and disagreement.
 - Citation: (Johnson and Gillingham 2005)
3. Empirical SDMs provide an alternative method of estimating habitat relationships directly from observed species occurrence.
 - Establish the conceptual transition from HSI approaches to MaxEnt.
 - Citations: (Elith and Leathwick 2009; Johnson and Gillingham 2005)

6. Maximum Entropy species distribution modeling

1. Introduce MaxEnt as a presence-background species distribution modeling framework.

- Particularly useful where reliable absence observations are unavailable.
 - Relates environmental conditions at known occurrence locations to environmental conditions available throughout the modeling region.
 - Citations: (Phillips, Anderson, and Schapire 2006; Elith et al. 2011)
2. Explain why MaxEnt is appropriate for Eastern oyster occurrence data in PBS.
 - FWC oyster-bed data provide known oyster occurrences without systematically sampled absence locations.
 - Presence-background modeling therefore fits the available data structure.
 - Citations: (Phillips, Anderson, and Schapire 2006; FWC-FWRI 2026)
 3. Explain important considerations when implementing MaxEnt.
 - Background extent and sampling.
 - Spatial sampling bias.
 - Predictor selection.
 - Regularization.
 - Feature classes and model complexity.
 - Citations: (Phillips et al. 2009; Merow, Smith, and Silander 2013; Elith et al. 2011)
 4. Explain the need for model tuning and spatial validation.
 - MaxEnt should not be treated as an automated black-box model.
 - Model configuration should be evaluated systematically.
 - Spatially independent validation should be used where occurrence data exhibit spatial structure.
 - Citations: (Kass et al. 2021; Roberts et al. 2017; Valavi et al. 2019)

7. Research rationale, objectives, and hypotheses

1. Establish the primary research problem.
 - An estuary-specific HSI already exists for Eastern oyster restoration within PBS.
 - This study will independently develop an empirically fitted MaxEnt habitat-suitability model using observed oyster occurrences and environmental conditions.
 - Citations: (Geselbracht et al. 2024; Phillips, Anderson, and Schapire 2006)
2. Primary objective.
 - Develop a spatially explicit MaxEnt SDM representing relative Eastern oyster habitat suitability throughout PBS.
3. Secondary objective.
 - Quantitatively compare the MaxEnt suitability surface with the existing Pensacola Bay HSI.
4. Supporting objectives.
 - Identify environmental predictors associated with observed oyster distribution.

- Evaluate MaxEnt predictive performance using spatially separated validation data.
 - Identify geographic areas where MaxEnt and the HSI agree or disagree.
 - Evaluate whether MaxEnt provides a reproducible empirical complement to HSI-based restoration planning.
 - Citations: (Johnson and Gillingham 2005; Geselbracht et al. 2024; Merow, Smith, and Silander 2013)
5. Research hypothesis.
- The MaxEnt model is expected to identify patterns of habitat suitability consistent with established Eastern oyster habitat requirements.
 - Measurable agreement is expected between MaxEnt and the existing HSI, with spatial disagreement arising from differences between empirically fitted environmental relationships and predefined HSI suitability relationships.
 - Citations: (Johnson and Gillingham 2005; Geselbracht et al. 2024)

Methods

1. Study design and team status

1. State project team status.
 - This capstone is being completed as an individual project.
 - All data acquisition, processing, modeling, analysis, and interpretation will be conducted by the author.
2. Describe the overall analytical workflow.
 - Environmental and occurrence data acquisition.
 - Data QA/QC and preprocessing.
 - Creation of continuous environmental predictor surfaces.
 - Oyster occurrence preparation.
 - Background sampling.
 - Predictor screening.
 - MaxEnt tuning and fitting.
 - Spatial model validation.
 - HSI comparison.
 - Citations: (Phillips, Anderson, and Schapire 2006; Merow, Smith, and Silander 2013; Kass et al. 2021)

2. Study area

1. Define the Pensacola Bay System modeling extent.
 - Delineate the study area using selected Florida Department of Environmental Protection Waterbody Identification (WBID) polygons.

- Describe the primary embayments included within the analysis.
 - Citations: (FDEP 2005; Geselbracht et al. 2024)
2. Restrict the modeling domain to aquatic habitat.
 - Background and prediction locations should represent environments potentially available to Eastern oysters.
 - Terrestrial areas should be excluded from the modeling domain.
 - Citations: (Phillips et al. 2009; Merow, Smith, and Silander 2013)
 3. Maintain spatial compatibility with the existing Pensacola Bay HSI where possible.
 - A comparable geographic extent will support subsequent spatial comparison.
 - Citation: (Geselbracht et al. 2024)
 4. Develop a study-area map.
 - PBS boundary.
 - Major embayments.
 - Oyster occurrence locations.
 - Environmental monitoring locations.

3. Data acquisition and dataset inventory

1. Eastern oyster occurrence data.
 - Source data from the FWC-FWRI Oyster Beds in Florida dataset.
 - Record dataset version and acquisition date.
 - Document original geometry type.
 - Document number of source features and number retained within the study area.
 - Document temporal/source-data information where available.
 - Citation: (FWC-FWRI 2026)
2. Water-quality observations.
 - Retrieve observations from the Water Quality Portal.
 - Use the `dataRetrieval` package and associated WQP services.
 - Restrict queries to relevant counties, estuarine monitoring locations, selected parameters, and the study period.
 - Citations: (USGS 2026; DeCicco et al. 2026)
3. Document water-quality dataset size and temporal coverage.
 - Dissolved oxygen: approximately 145 stations and 3,072 observations.
 - Salinity: approximately 135 stations and 3,401 observations.
 - Temperature: approximately 145 stations and 3,442 observations.
 - Turbidity: approximately 122 stations and 1,563 observations.
 - Chlorophyll-a: approximately 48 stations and 754 observations.
 - Current temporal coverage is approximately 2005–2024.
 - Citations for source data: (USGS 2026; DeCicco et al. 2026)
4. Evaluate candidate predictors based on spatial coverage.

- Chlorophyll-a is currently expected to be excluded from interpolation because available stations do not provide adequate system-wide spatial coverage.
 - Salinity, dissolved oxygen, temperature, and turbidity currently provide substantially stronger monitoring coverage.
 - Ecological rationale for candidate variables: (Peyre, Marshall, and Sable 2021; Geselbracht et al. 2024)
5. Bathymetry.
 - Obtain Pensacola Bay estuarine bathymetry from NOAA.
 - Document raster resolution, vertical datum, geographic extent, and preprocessing requirements.
 - Citation: (National Centers for Environmental Information 2018)
 6. Substrate or bottom type.
 - Identify and acquire a suitable spatial dataset describing bottom characteristics.
 - Document spatial resolution, classes, temporal source, and coverage.
 - Ecological justification: (Barnes et al. 2007; Geselbracht et al. 2024)
 - **Citation needed for final substrate dataset.**
 7. Existing Pensacola Bay HSI.
 - Obtain the original GIS/raster output used by Geselbracht et al.
 - Document raster score range, projection, extent, resolution, and NoData handling.
 - Citation for methodology: (Geselbracht et al. 2024)
 - **Add a separate dataset citation once the original HSI GIS product is acquired.**

4. Water-quality preprocessing

1. Standardize measurement characteristics and units.
 - Avoid combining fundamentally different measurement types that share similar characteristic names.
 - Retain comparable units and analytical forms for each predictor.
 - Citations: (USGS 2026; DeCicco et al. 2026)
2. Restrict observations to warm-season conditions.
 - Current working definition: May through October.
 - Use seasonal filtering to characterize environmental conditions relevant to warm-season oyster stress and habitat suitability.
 - Citation for Pensacola seasonal precedent: (Geselbracht et al. 2024)
3. Generate station-level environmental summaries.
 - Calculate warm-season mean.
 - Calculate standard deviation.
 - Record total observation count.
 - Record number of unique years represented.

- Record first and last sample date.
 - Consider calculating site-year warm-season means prior to calculating long-term site means to avoid over-weighting heavily sampled individual years.
4. Perform QA/QC.
 - Identify potential unit inconsistencies.
 - Investigate implausible or extreme values.
 - Document removal or correction criteria.
 - Retain original raw data separately from processed datasets.
 5. Convert monitoring data to spatial features.
 - Use `sf` for spatial-vector processing.
 - Project observations to NAD 1983 (2011) StatePlane Florida North FIPS 0903 in meters.
 - Citation: (Pebesma 2018)

5. Generation of continuous environmental predictor surfaces

1. Assess monitoring-station spatial coverage prior to interpolation.
 - Map station distribution for each parameter.
 - Identify spatial clustering.
 - Identify portions of PBS lacking observations.
 - Exclude variables where prediction would consist primarily of spatial extrapolation.
2. Account for coastal barriers and complex estuarine geometry.
 - Straight-line Euclidean distance may incorrectly represent spatial relationships between monitoring locations separated by peninsulas, islands, or narrow water connections.
 - Barrier-aware interpolation will be evaluated where feasible.
 - **Suggested citation: Stachelek and Madden (2015).**
3. Evaluate inverse path distance weighting (IPDW) as a candidate interpolation method.
 - IPDW uses aquatic path distance rather than straight-line Euclidean distance.
 - This approach is specifically suited to coastal water-quality interpolation where land barriers affect connectivity.
 - **Suggested citation: Stachelek and Madden (2015).**
4. Compare candidate interpolation approaches.
 - Potential methods include:
 - Inverse path distance weighting.
 - Conventional inverse distance weighting.
 - Ordinary kriging.
 - Use spatially withheld monitoring locations to evaluate predictive performance.
 - Compare RMSE, MAE, and bias.

- Select the final interpolation method based on predictive performance and ecological appropriateness.
 - Citations: (Roberts et al. 2017; Valavi et al. 2019)
5. Produce final environmental predictor rasters.
 - Use a common projected coordinate system.
 - Use a common cell size.
 - Use common raster origin and alignment.
 - Mask predictions to aquatic habitat.
 - Document final raster resolution.
 - Citation for Pensacola spatial-modeling precedent: (Geselbracht et al. 2024)

6. Oyster occurrence preparation

1. Clip FWC oyster-bed features to the study area.
 - Remove occurrences outside the PBS modeling extent.
 - Citation: (FWC-FWRI 2026)
2. Convert mapped oyster-bed polygons into representative MaxEnt presence observations.
 - Use a reproducible procedure to avoid overrepresenting large or heavily mapped oyster polygons.
 - Document the initial number of oyster polygons and final number of presence observations.
 - Citations: (Phillips et al. 2009; Merow, Smith, and Silander 2013)
3. Evaluate spatial clustering of presence observations.
 - Apply spatial thinning where necessary.
 - Thinning should be based on sampling bias and spatial dependence rather than arbitrary data reduction.
 - Citations: (Phillips et al. 2009; Merow, Smith, and Silander 2013)

7. Background sampling

1. Define the accessible/background modeling region.
 - Restrict background sampling to aquatic habitat within PBS.
 - Require complete environmental predictor coverage.
 - Exclude terrestrial areas and locations outside the modeled environmental domain.
 - Citations: (Phillips et al. 2009; Merow, Smith, and Silander 2013)
2. Generate MaxEnt background locations.
 - Document the number of background points.
 - Sample from the defined accessible region rather than an arbitrary rectangular extent.
 - Citations: (Phillips et al. 2009; Merow, Smith, and Silander 2013)

3. Address occurrence-sampling bias.
 - Evaluate whether spatial thinning, bias weighting, or restricted background sampling is necessary.
 - Citation: (Phillips et al. 2009)

8. Predictor screening

1. Assemble final candidate predictor stack.
 - Salinity.
 - Dissolved oxygen.
 - Temperature.
 - Turbidity.
 - Bathymetry/depth.
 - Substrate/bottom type, if suitable data are obtained.
 - Citations: (Peyre, Marshall, and Sable 2021; Barnes et al. 2007; Pollack et al. 2012; Geselbracht et al. 2024)
2. Evaluate correlation and redundancy among predictors.
 - Calculate pairwise correlations.
 - Identify highly correlated environmental variables.
 - Retain predictors based on ecological relevance, data quality, and interpretability.
 - **Suggested citation: Dormann et al. (2013).**

9. MaxEnt model fitting and tuning

1. Fit MaxEnt using occurrence locations, background samples, and the finalized environmental predictor stack.
 - Citations: (Phillips, Anderson, and Schapire 2006; Elith et al. 2011)
2. Tune model complexity rather than relying exclusively on default settings.
 - Evaluate multiple regularization multipliers.
 - Evaluate multiple feature-class combinations.
 - Citations: (Merow, Smith, and Silander 2013; Kass et al. 2021)
3. Use the **ENMeval** framework for model tuning and evaluation.
 - Citation: (Kass et al. 2021)
4. Select the final MaxEnt configuration.
 - Consider predictive performance.
 - Consider omission rates.
 - Consider model complexity.
 - Citations: (Kass et al. 2021; Merow, Smith, and Silander 2013)
5. Generate the final continuous habitat-suitability surface.
 - Retain continuous predictions for primary comparison with the HSI.
 - Citations: (Elith et al. 2011; Merow, Smith, and Silander 2013)

6. Evaluate environmental-variable influence.
 - Examine permutation importance.
 - Examine response curves.
 - Evaluate other supported measures of predictor contribution.
 - Interpret associations as predictive relationships rather than direct causal effects.
 - Citations: (Elith et al. 2011; Merow, Smith, and Silander 2013)

10. Spatial model validation

1. Use spatial rather than purely random cross-validation.
 - Nearby oyster occurrences may experience highly similar environmental conditions.
 - Random partitions may therefore produce overly optimistic estimates of predictive performance.
 - Citations: (Roberts et al. 2017; Valavi et al. 2019)
2. Divide oyster occurrences into spatially separated validation folds.
 - Select spatial-block dimensions based on observed spatial structure.
 - Citation: (Valavi et al. 2019)
3. Evaluate models against withheld spatial folds.
 - Report appropriate presence-background performance metrics.
 - Consider omission rate and AUC where appropriate.
 - Report variation in model performance among folds.
 - Citations: (Phillips, Anderson, and Schapire 2006; Kass et al. 2021)

11. Comparison with the existing Habitat Suitability Index

1. Harmonize the MaxEnt and HSI raster surfaces.
 - Match geographic extent.
 - Match coordinate reference system.
 - Match cell size and alignment.
 - Ensure suitability values are directly comparable.
 - Citations: (Geselbracht et al. 2024; Johnson and Gillingham 2005)
2. Compare continuous habitat-suitability predictions.
 - Calculate cell-by-cell correlation.
 - Consider Pearson and/or Spearman correlation depending on model-output distributions.
 - Citation: (Johnson and Gillingham 2005)
3. Generate a spatial difference surface.

$$D_i = S_{MaxEnt,i} - S_{HSI,i}$$

- Values near zero represent agreement.
 - Positive values identify locations where MaxEnt predicts relatively greater suitability.
 - Negative values identify locations where the HSI predicts relatively greater suitability.
 - Citation: (Johnson and Gillingham 2005)
4. Evaluate categorical agreement if suitability classes are created.
 - Both models predict high suitability.
 - Both models predict low suitability.
 - MaxEnt predicts high suitability while HSI predicts low suitability.
 - HSI predicts high suitability while MaxEnt predicts low suitability.
 - Consider an agreement statistic if appropriate.
 - Citation: (Johnson and Gillingham 2005)
 5. Interpret disagreement between models.
 - Determine whether disagreement is geographically concentrated.
 - Evaluate whether disagreement corresponds with particular environmental predictors.
 - Consider differences in data availability and modeling assumptions.
 - Avoid treating either model as automatically representing ground truth.
 - Citations: (Johnson and Gillingham 2005; Geselbracht et al. 2024)

12. Software and reproducibility

1. Conduct data processing, statistical analysis, and modeling in R.
 - Citation: (R Core Team 2024)
2. Use scripted workflows for data transformation and spatial processing.
 - `tidyverse` for data manipulation.
 - `sf` for vector spatial processing.
 - Citations: (Wickham et al. 2019; Pebesma 2018)
3. Maintain reproducibility throughout the workflow.
 - Record package versions.
 - Set random seeds where applicable.
 - Record data-acquisition dates.
 - Document model settings and tuning parameters.
 - Preserve raw data separately from processed data.
 - Citations for dynamic data sources: (DeCicco et al. 2026; USGS 2026; FWC-FWRI 2026)

Additional References to Obtain

1. Barrier-aware interpolation

- Stachelek, J., & Madden, C. J. (2015). *Application of inverse path distance weighting for high-density spatial mapping of coastal water quality patterns. International Journal of Geographical Information Science*, 29(7), 1240–1250.
- DOI: 10.1080/13658816.2015.1018833
- Use for justification of inverse path distance weighting and coastal-barrier-aware interpolation.

2. Predictor collinearity

- Dormann, C. F., Elith, J., Bacher, S., et al. (2013). *Collinearity: a review of methods to deal with it and a simulation study evaluating their performance. Ecography*, 36(1), 27–46.
- DOI: 10.1111/j.1600-0587.2012.07348.x
- Use for predictor screening and treatment of correlated environmental variables.

3. IPDW R package

- Add the current package citation if `ipdw` is used for the final interpolation workflow.
- DOI: 10.32614/CRAN.package.ipdw

- Barnes, T. K., A. K. Volety, K. Chartier, F. J. Mazzotti, and L. Pearlstine. 2007. “A Habitat Suitability Index Model for the Eastern Oyster (*Crassostrea virginica*), a Tool for Restoration of the Caloosahatchee Estuary, Florida.” *Journal of Shellfish Research* 26 (4): 949–59. [https://doi.org/10.2983/0730-8000\(2007\)26%5B949:AHSIMF%5D2.0.CO;2](https://doi.org/10.2983/0730-8000(2007)26%5B949:AHSIMF%5D2.0.CO;2).
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- FWC-FWRI. 2026. “Oyster Beds in Florida.” Geospatial Open Data; Florida Fish; Wildlife Conservation Commission, Fish; Wildlife Research Institute. <https://geodata.myfwc.com/datasets/myfwc::oyster-beds-in-florida/about>.

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- Grabowski, Jonathan H., Robert D. Brumbaugh, Robert F. Conrad, Andrew G. Keeler, James J. Opaluch, Charles H. Peterson, Michael F. Piehler, Sean P. Powers, and Ashley R. Smyth. 2012. “Economic Valuation of Ecosystem Services Provided by Oyster Reefs.” *BioScience* 62 (October): 900–909. <https://doi.org/10.1525/bio.2012.62.10.10>.
- Johnson, Chris J., and Michael P. Gillingham. 2005. “An Evaluation of Mapped Species Distribution Models Used for Conservation Planning.” *Environmental Conservation* 32 (June): 117–28. <https://doi.org/10.1017/S0376892905002171>.
- Kass, Jamie M., Robert Muscarella, Peter J. Galante, Corentin L. Bohl, Gonzalo E. Pinilla-Buitrago, Robert A. Boria, Mariano Soley-Guardia, and Robert P. Anderson. 2021. “ENMeval 2.0: Redesigned for Customizable and Reproducible Modeling of Species’ Niches and Distributions.” *Methods in Ecology and Evolution* 12 (September): 1602–8. <https://doi.org/10.1111/2041-210X.13628>.
- Merow, Cory, Matthew J. Smith, and John A. Silander. 2013. “A Practical Guide to MaxEnt for Modeling Species’ Distributions: What It Does, and Why Inputs and Settings Matter.” *Ecography* 36 (October): 1058–69. <https://doi.org/10.1111/j.1600-0587.2013.07872.x>.
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Wickham, Hadley, Mara Averick, Jennifer Bryan, Winston Chang, Lucy D’Agostino McGowan, Romain François, Garrett Grolemond, et al. 2019. “Welcome to the tidyverse.” *Journal of Open Source Software* 4 (43): 1686. <https://doi.org/10.21105/joss.01686>.